

Improving the Sequence Contribution in PolyClip

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To improve the contribution of the sequence modality within the PolyClip framework, we designed and evaluated three different training configurations on a subset of 5,000 variants without changing the validation set and 200 epochs. The objective of these experiments was to investigate how the distribution of information between the sequence branch and the biological feature branch influences the quality of the learned multimodal representations, particularly for the study of complex and polygenic diseases.

The first configuration corresponded to the original PolyClip architecture. In this baseline model, the sequence branch mainly relied on genomic sequence representations derived from REF/ALT or Allele1/Allele2-centered nucleotide windows and contextual embeddings, while the biological branch contained most of the structured annotation. In this setting, biological and ontological features contributed strongly to the organization of the latent space, whereas the sequence branch showed weaker discriminative power.

To strengthen the sequence encoder, a second model was constructed by transferring several biologically informative annotation columns from the biological modality into the sequence modality. These columns were specifically selected because they describe the functional and structural impact of genomic variants and therefore provide complementary information that can enhance sequence interpretation. The transferred columns were:

- IMPACT
- VARIANT_CLASS
- STRAND
- Distance_Grantham
- SIFT_bin
- PolyPhen_ord
- Impact_num
- VariantClass_num

These features were integrated into the sequence branch because they directly characterize the predicted biological effect of a variant at the protein and functional level. For example, **IMPACT** and **Variant_Class** provide descriptions of variant severity and mutation type, while **SIFT_bin** and **PolyPhen_ord** contain functional pathogenicity predictions derived from established bioinformatics tools. Similarly, **Distance_Grantham** quantifies the biochemical difference between amino acids, offering additional information regarding the potential structural consequences of amino acid substitutions.

In the second configuration, these features were removed from the biological branch and incorporated exclusively into the sequence representation pipeline. The objective was to encourage the sequence encoder to learn biologically enriched sequence representations instead of relying solely on nucleotide context information.

Finally, a third configuration was evaluated in which the same sequence-effect columns were retained simultaneously in both the sequence branch and the biological branch. Rather than redistributing the information between modalities, this approach introduced a controlled redundancy across modalities, allowing both branches to access the same functional and pathogenic annotations. The goal of this configuration was to determine whether maintaining shared information across modalities could improve multimodal alignment and stabilize the fused representation space.

The comparative analysis of these three models revealed substantial differences in latent space organization and retrieval behavior. The baseline PolyClip model showed strong dependence on biological annotations, with the sequence branch contributing less effectively to variant discrimination. In the second configuration, enriching the sequence branch with functional effect annotations improved the ability of the sequence encoder to capture biologically meaningful patterns and increased the contribution of sequence embeddings within the fused latent space.